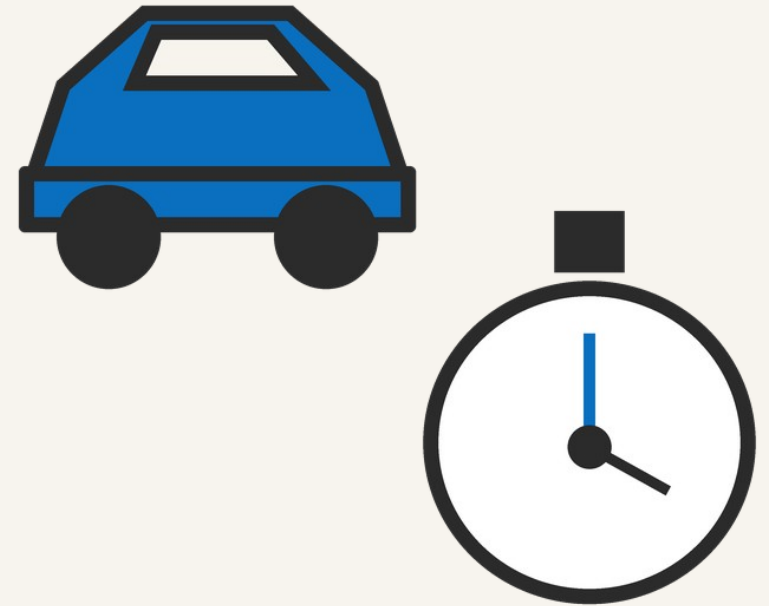


OUR DYNAMIC UNIVERSE 1 • MOTION

Motion, Graphs & Projectiles

The equations, the graphs, and what happens when you throw something.

Higher Physics



Starter

2008 Higher Physics

Scan the code and answer these:

Q8

Q10

Q18

Section A · particles, semiconductors

Show your working, not just the letter.



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PART 1

Describing motion

Every quantity in this course is one of two kinds — knowing which matters more than it looks like it should.

By the end of this part...

- 1 I can state the difference between a scalar and a vector quantity
- 2 I can state the difference between distance and displacement, and between speed and velocity
- 3 I can calculate average speed and velocity for a journey

Scalars and vectors

- A scalar quantity has magnitude (size) only – e.g. distance, speed, time, mass.
- A vector quantity has magnitude AND direction – e.g. displacement, velocity, acceleration, force.
- The same journey can have a scalar answer and a vector answer that are completely different numbers.

Distance and displacement

DISTANCE (scalar)

- Total length of path travelled
- Always positive, always adds up
- Unit: metre (m)

DISPLACEMENT (vector)

- Straight-line distance start to finish, in a stated direction
- Can be zero even if distance travelled is large
- Unit: metre (m)

Worked example – distance vs displacement

A runner jogs 400 m round a track and finishes back at the start.
Find the distance and the displacement for the lap.

$$d = ?$$

$$s = ?$$

$$d = 400 \text{ m}$$

$$s = 0 \text{ m}$$

(back where she started)

Speed and velocity

$$d/t$$

d

distance

t

time

Average velocity = displacement / time, in the direction of the displacement — speed is how fast; velocity is how fast AND which way.

Watch Out – velocity is a vector

THE MISTAKE

- Giving a speed when the question asks for velocity
- Not stating a direction with a velocity answer

THE CORRECT ANSWER

- A velocity answer needs a direction, not just a number.
- "12 m s⁻¹ north" is complete; "12 m s⁻¹" alone is not.



Scalars, vectors, and speed vs velocity

Scalar: magnitude only (distance, speed, time, mass).

Vector: magnitude AND direction (displacement, velocity, acceleration, force).

average speed = distance/time

average velocity = displacement/time

Your turn – describing motion

SHOW ME BOARDS

- 1 State whether mass, force, and time are scalars or vectors.
- 2 A car drives 3 km east then 3 km west. State its distance travelled and its displacement.
- 3 A cyclist covers 500 m in 40 s. Calculate the average speed.

TRY THIS – Answers

- 1 Mass is a scalar; force and time – force is a vector, time is a scalar.
- 2 Distance = 6 km; displacement = 0 (back at the start).
- 3 12.5 m s^{-1} .

Starter

2011 Higher Physics

Scan the code and answer these:

Q1

Q8

Q18

Section A · vectors, semiconductors
Show your working, not just the letter.



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PART 2

The equations of motion

Four relationships link every straight-line, constant-acceleration problem you will ever be asked.

By the end of this part...

- 1 I can define acceleration and calculate it from a change in velocity
- 2 I can select and use the four equations of motion
- 3 I can describe an experiment to measure the acceleration of an object down a slope

Acceleration

$$a = v - u / t$$

a

acceleration
(m s⁻²)

v

final velocity
(m s⁻¹)

u

start velocity
(m s⁻¹)

t

time (s)

Acceleration is a vector: a positive value speeds an object up in the positive direction; a negative value slows it down.

The equations of motion

1 $s = ((u + v) / 2) t$

2 $v = u + at$

3 $s = ut + \frac{1}{2}at^2$

4 $v^2 = u^2 + 2as$



The equations of motion – 1 of 2

For an object moving with **CONSTANT** acceleration in a straight line:

$$s = u + v/2 t$$

$$v = u + at$$



The equations of motion – 2 of 2

$$s = ut + \frac{1}{2} a t^2$$

$$v^2 = u^2 + 2as$$

Worked example – the equations of motion

A car accelerates from rest at 2.5 m s^{-2} for 6.0 s . Find v and s .

$$u = 0 \text{ m s}^{-1}$$

$$a = 2.5 \text{ m s}^{-2}$$

$$t = 6.0 \text{ s}$$

$$v, s = ?$$

$$v = u + at$$

$$v = 0 + 2.5 \times 6.0$$

$$v = 15 \text{ m s}^{-1}$$

$$s = ut + \frac{1}{2} at^2$$

$$s = 0 + 0.5 \times 2.5 \times 6.0^2$$

$$\underline{s = 45 \text{ m}}$$



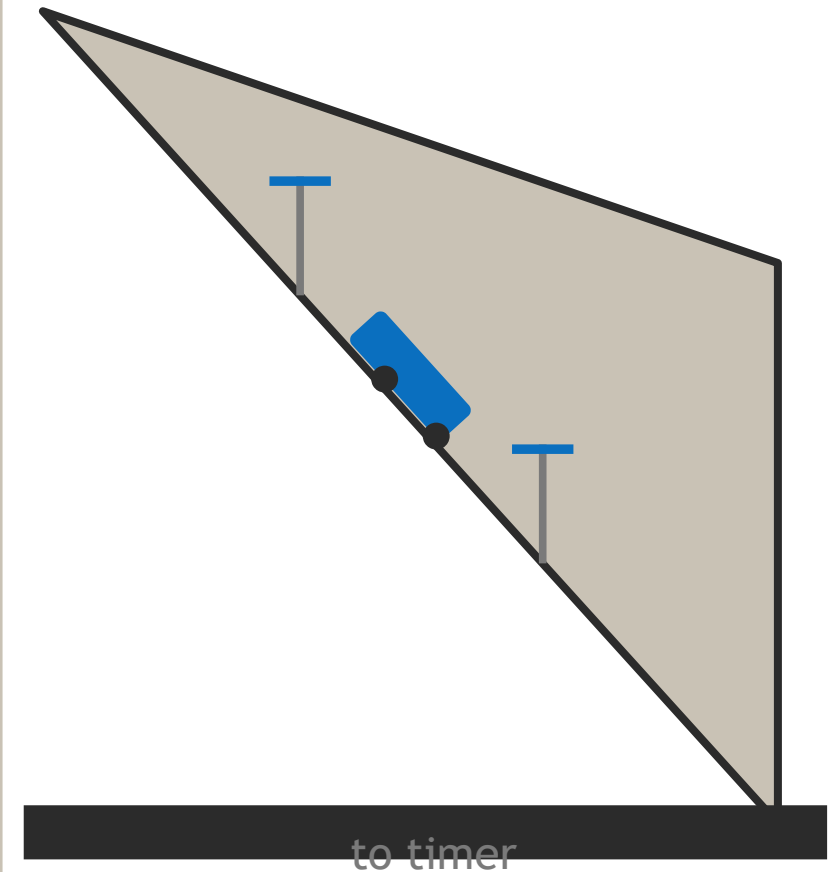
Measuring acceleration down a slope

A trolley is released from rest at the top of the slope.

Two light gates are set a fixed distance apart on the slope.

Velocity is measured at each gate; time is recorded between them.

$$a = \frac{v_2 - v_1}{t}$$



Watch Out — signs and units

THE MISTAKE

- Mixing up which quantity is u and which is v
- Treating a deceleration's negative sign as a mistake

THE CORRECT ANSWER

- u is the START velocity, v is the END velocity.
- A negative a means slowing down — that IS the answer.

Your turn – equations of motion

SHOW ME BOARDS

- 1 A cyclist accelerates from 2.0 m s^{-1} to 8.0 m s^{-1} in 3.0 s . Calculate the acceleration.
- 2 A trolley starts at rest and accelerates at 1.5 m s^{-2} for 4.0 s . Calculate the distance it travels.
- 3 State what two measurements the acceleration-down-a-slope experiment needs at each light gate.

TRY THIS – Answers

- 1 2.0 m s^{-2} .
- 2 12 m.
- 3 The velocity at that gate, and the time between the two gates.

Starter

2012 Higher Physics

Scan the code and answer these:

Q1

Q2

Q11

Section A · vectors, graphs, capacitors
Show your working, not just the letter.



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PART 3

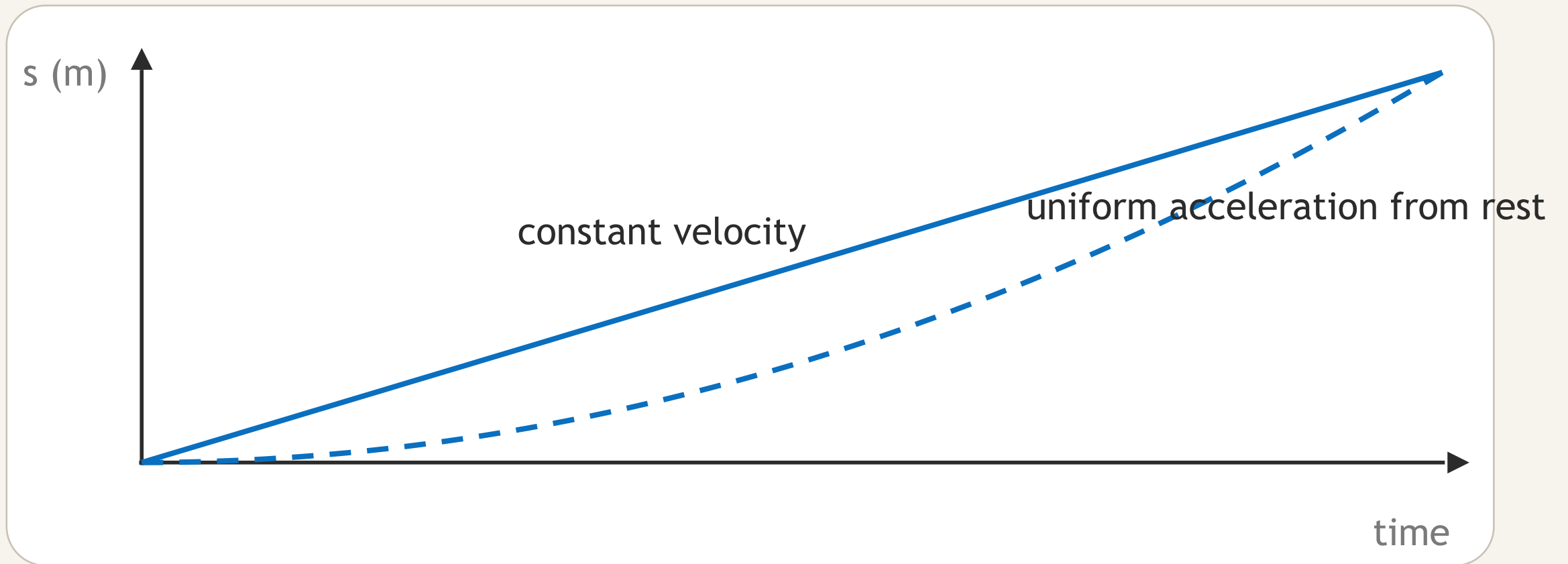
Motion-time graphs

Displacement, velocity and acceleration graphs are three views of the same motion — and you can read any one off the others.

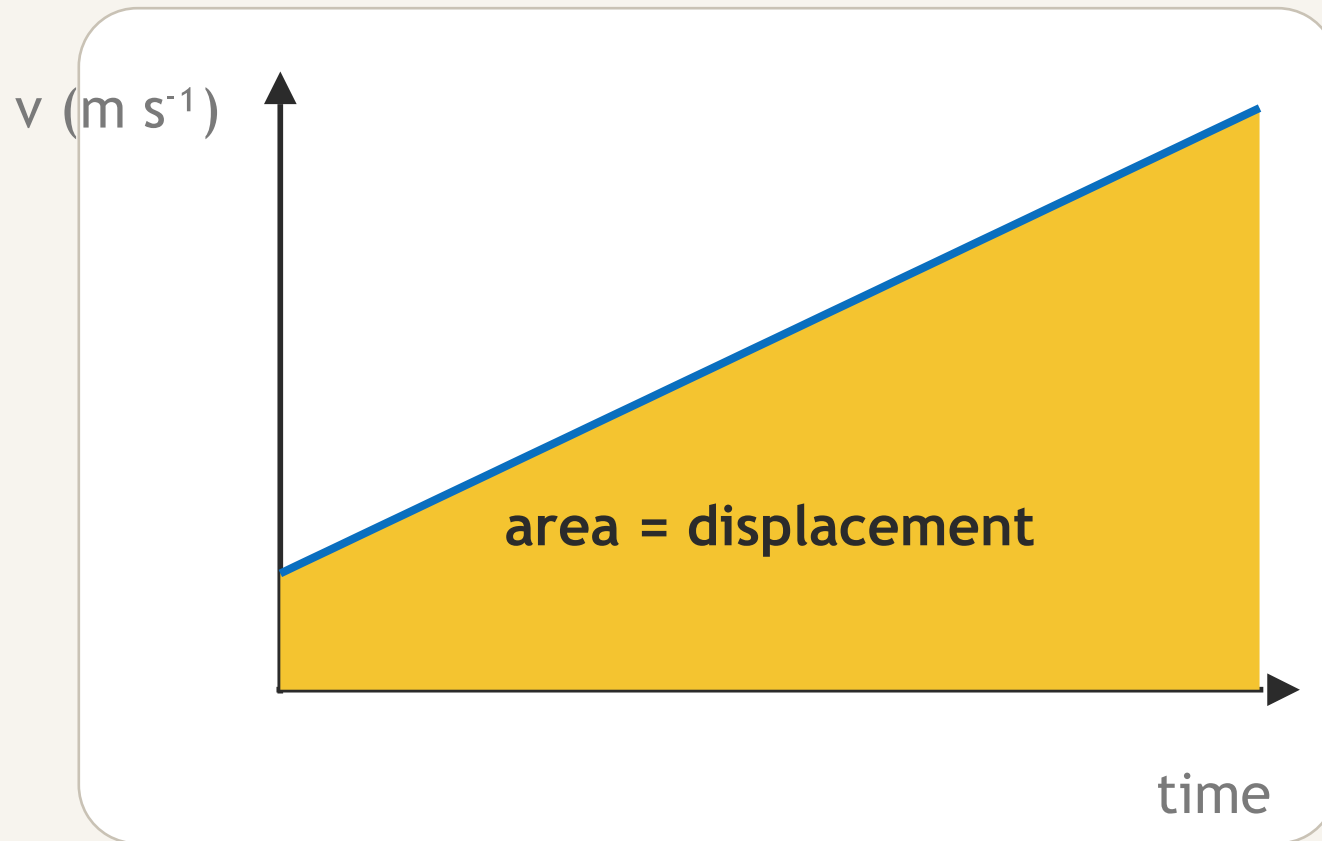
By the end of this part...

- 1 I can sketch and interpret displacement-time, velocity-time and acceleration-time graphs
- 2 I can calculate distance, displacement, velocity and acceleration from a graph
- 3 I can relate the three types of graph to each other

Displacement-time graphs



Velocity-time graphs

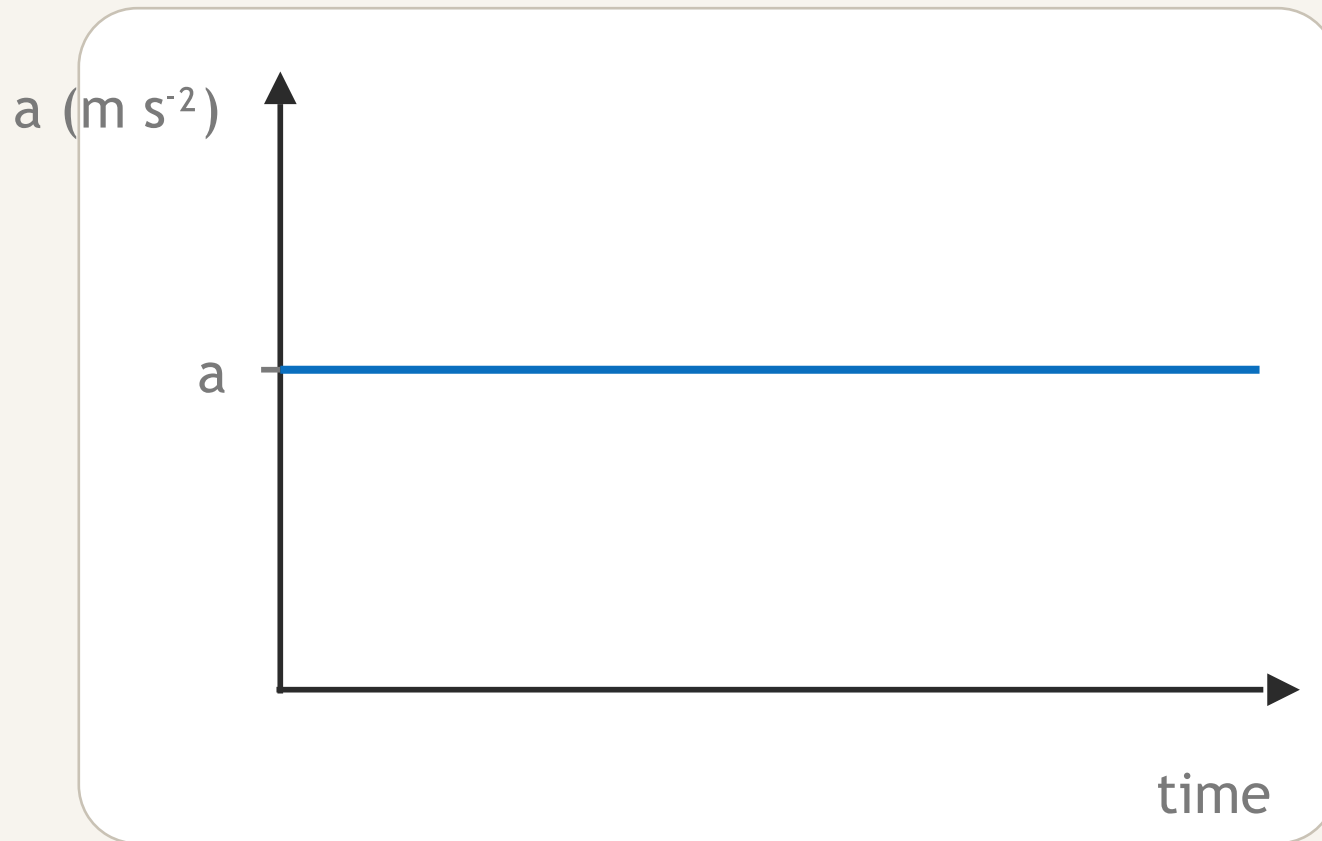


The gradient of a velocity-time graph is the acceleration.

The area under a velocity-time graph is the displacement.

A straight line means constant acceleration.

Acceleration-time graphs

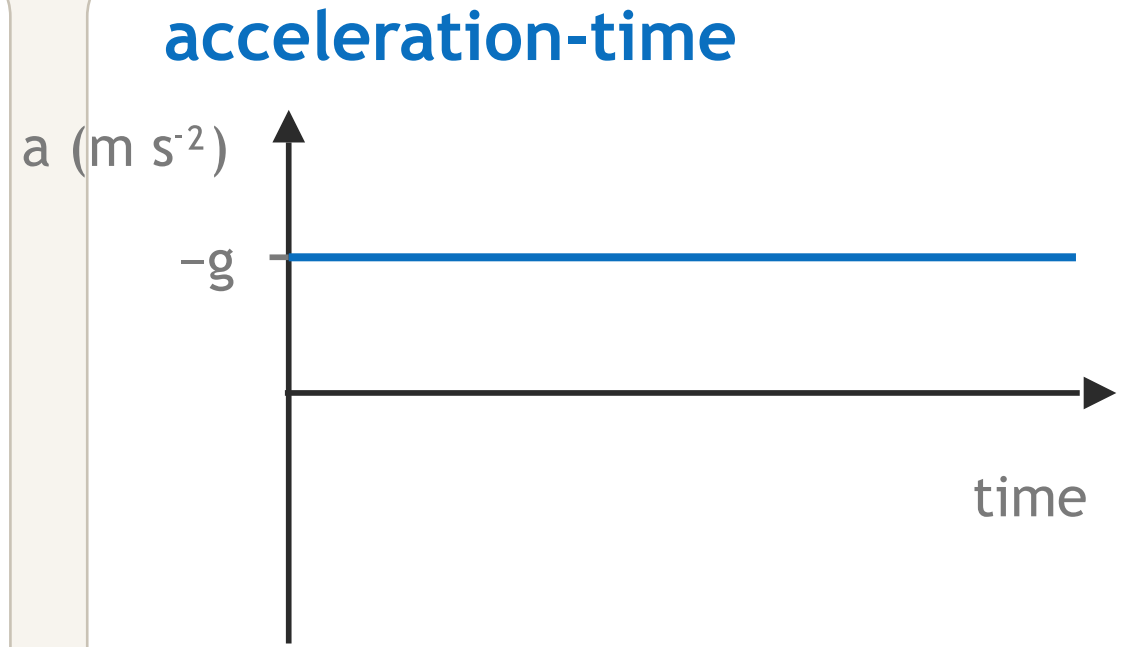
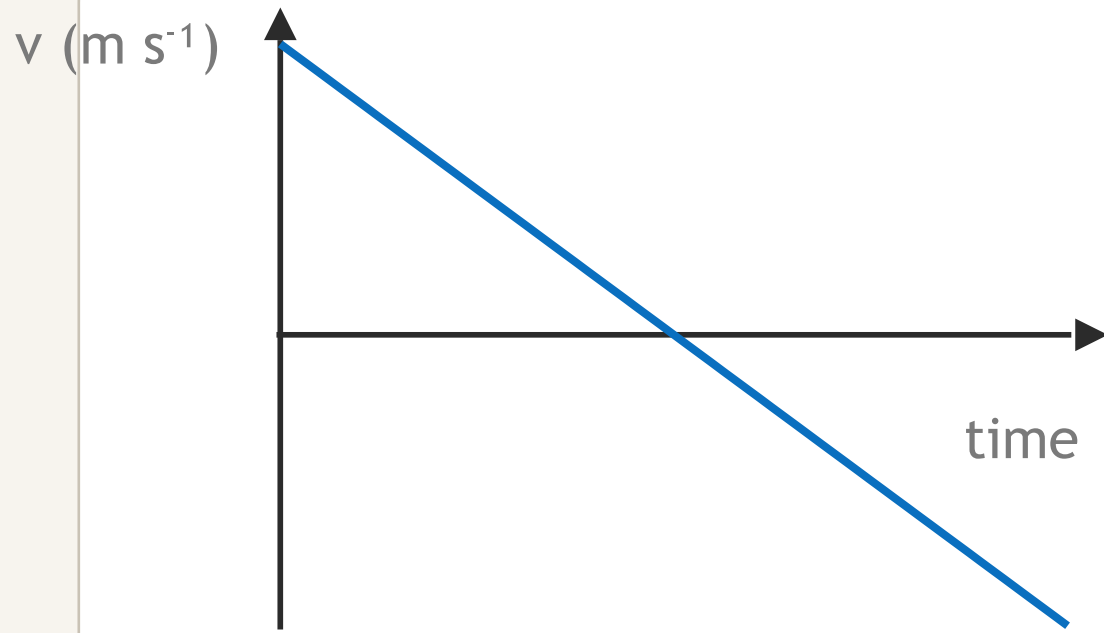


A flat line means constant acceleration.

The area under an acceleration-time graph is the change in velocity.

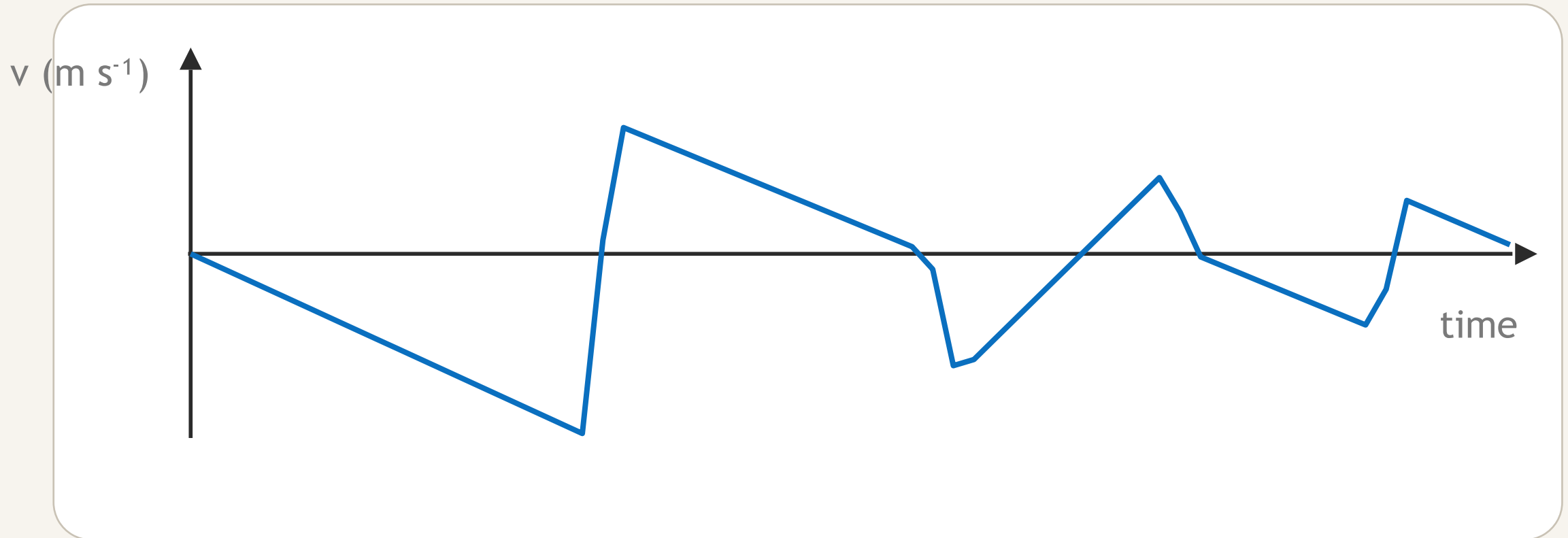
A flat line AT ZERO means constant velocity – no acceleration.

Thrown vertically upward



A bouncing ball

Taking upward as positive — each bounce loses some speed



Worked example – reading a velocity-time graph

A trolley's velocity-time graph rises from 0 to 8.0 m s^{-1} over 4.0 s , then stays constant at 8.0 m s^{-1} until $t = 10.0 \text{ s}$.

Find the acceleration over the first 4.0 s and the total distance.

$$\Delta v = 8.0 \text{ m s}^{-1}$$

$$\Delta t_1 = 4.0 \text{ s}$$

$$t_{\text{total}} = 10.0 \text{ s}$$

$$a = 8.0/4.0$$

$$a = 2.0 \text{ m s}^{-2}$$

$$s = \frac{1}{2} \times 4.0 \times 8.0 + 6.0 \times 8.0$$

$$s = 16 + 48$$

$$\underline{s = 64 \text{ m}}$$

Watch Out — matching graph to shape

THE MISTAKE

- Picking a velocity-time shape for an acceleration graph
- A straight d-t line for an accelerating object

THE CORRECT ANSWER

- Constant acceleration gives a flat a-t line, not sloped.
- d-t is a curve whenever the object accelerates.

Your turn – motion graphs

SHOW ME BOARDS

- 1 Sketch the velocity-time graph for an object released from rest and falling freely for 3 s.
- 2 A velocity-time graph is a horizontal line at 6 m s^{-1} for 5 s. Find the distance travelled and the acceleration.
- 3 State what is constant when a displacement-time graph is a straight line.

TRY THIS – Answers

- 1 A straight line through the origin, gradient $-g$ (steepening downward speed).
- 2 30 m; acceleration = 0.
- 3 Velocity.

Starter

2015 Higher Physics

Scan the code and answer these:

Q1

Q2

Q12

Section A · vectors, motion, a.c.

Show your working, not just the letter.



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PART 4

Projectiles — independent motions

A thrown ball is really two motions at once, and they never interfere with each other.

By the end of this part...

- 1 I can state that the horizontal and vertical motions of a projectile are independent
- 2 I can state that satellites are in free fall around a planet or star
- 3 I can resolve an initial velocity into horizontal and vertical components

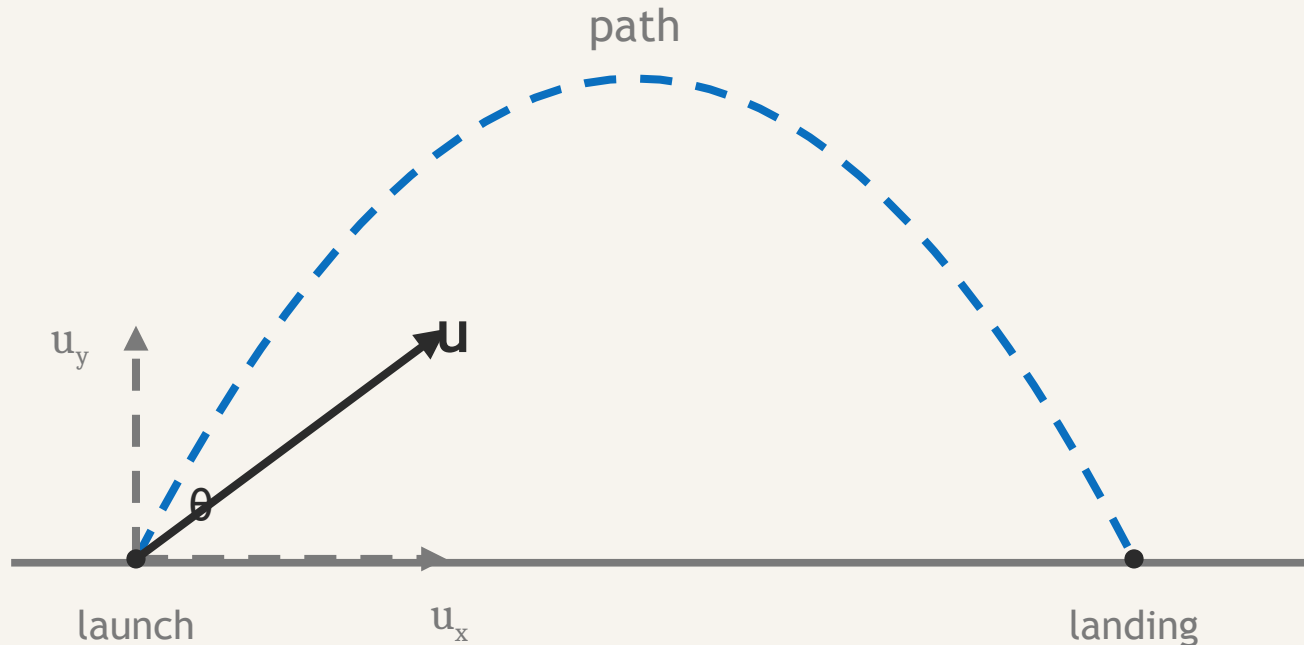
Independent motions

- A projectile's horizontal motion and vertical motion happen completely independently of each other.
- Horizontally: no force acts (ignoring air resistance), so horizontal velocity stays constant.
- Vertically: gravity acts, so vertical velocity accelerates at $g = 9.8 \text{ m s}^{-2}$, exactly as if the object were dropped.

Satellites — continuous free fall

- A satellite in orbit is falling towards the planet the whole time — it is in free fall.
- Its horizontal velocity is so large that, as it falls, the curved surface of the planet "falls away" beneath it at the same rate.
- It never gets any closer to the surface, so it keeps orbiting instead of landing.

Resolving the initial velocity



$$u_x = u \cos \theta$$

$$u_y = u \sin \theta$$

u is the launch speed and θ the angle above the horizontal. u_x stays constant throughout the flight; u_y changes under gravity.



Projectiles – the two independent motions

Horizontal: constant velocity (no horizontal force).

Vertical: constant acceleration $g = 9.8 \text{ m s}^{-2}$ (as if dropped).

$$u_x = u \cos \theta$$

$$u_y = u \sin \theta$$

A satellite orbits because it is in continuous free fall around the planet.

Watch Out — the vector nature of velocity

THE MISTAKE

- Using the launch speed for the whole flight
- Not resolving an angled launch first

THE CORRECT ANSWER

- Only u_x stays constant — u itself does not.
- Resolve first: use u_x and u_y , never u .

Your turn – projectile theory

SHOW ME BOARDS

- 1 State the size and direction of the horizontal force on a projectile (ignoring air resistance).
- 2 A ball is launched at 15 m s^{-1} at 40° above the horizontal. Calculate its horizontal and vertical velocity components.
- 3 Explain why a satellite in orbit does not fall out of the sky.

TRY THIS – Answers

- 1 Zero — there is no horizontal force.
- 2 $u_x = 11.5 \text{ m s}^{-1}$; $u_y = 9.64 \text{ m s}^{-1}$.
- 3 It is falling continuously, but its horizontal speed means the curved ground falls away beneath it at the same rate.

Starter

2014 Revised Higher Physics

Scan the code and answer these:

Q1

Q9

Q19

Section A · graphs, charged particles
Show your working, not just the letter.



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PART 5

Projectile calculations

Same two motions, solved separately, then combined at the end.

By the end of this part...

- 1 I can calculate the time of flight, range and height of a projectile
- 2 I can solve a projectile problem for a horizontal launch and for an angled launch

Worked example – horizontal launch

A ball rolls off a table 1.25 m high, horizontal speed 3.0 m s⁻¹.
Find the time to fall, and the horizontal distance travelled.

$$s_v = 1.25 \text{ m}$$

$$u_v = 0 \text{ m s}^{-1}$$

$$a = 9.8 \text{ m s}^{-2}$$

$$u_x = 3.0 \text{ m s}^{-1}$$

$$s = ut + \frac{1}{2} at^2$$

$$1.25 = 0.5 \times 9.8 \times t^2$$

$$t = 0.51 \text{ s}$$

$$s_x = u_x t$$

$$s_x = 3.0 \times 0.51$$

$$\underline{s_x = 1.5 \text{ m}}$$

Worked example – angled launch

A ball is launched at 20 m s^{-1} , 30° above horizontal, level ground.
Find the time to maximum height, and the horizontal range.

$$u = 20 \text{ m s}^{-1}$$

$$\theta = 30^\circ$$

$$u_y = 10 \text{ m s}^{-1}$$

$$u_x = 17.3 \text{ m s}^{-1}$$

$$\text{at max height, } v_y = 0$$

$$v_y = u_y - gt$$

$$t_{\text{up}} = 1.0 \text{ s} \implies t_{\text{total}} = 2.0 \text{ s}$$

$$s_x = u_x t$$

$$\underline{s_x = 17.3 \times 2.0 = 35 \text{ m}}$$

Watch Out — sign conventions

THE MISTAKE

- Mixing signs between rising and falling
- Using u instead of u_x for range

THE CORRECT ANSWER

- Pick one positive direction and keep it throughout.
- Range always uses u_x and the TOTAL flight time.

Checkpoint

- 1 State the difference between a scalar and a vector quantity, with one example of each.
- 2 Write down the four equations of motion.
- 3 Sketch the velocity-time graph for an object thrown vertically upward and caught again.
- 4 State what is true about the horizontal and vertical motions of a projectile.

Exam Practice

2023 Q1 – scalars/vectors and displacement.

2024 Q1 – reading a displacement-time graph.

2024 Q1(a) – projectile components and horizontal distance.

Past papers, Section A

LOOKING AHEAD

Next: Forces, Energy & Power

What makes an object accelerate in the first place — and what it costs to make it happen.

Plenary

- 1 State the difference between distance and displacement.
- 2 Write the equation linking acceleration, change in velocity and time.
- 3 State what the gradient and the area under a velocity-time graph each represent.
- 4 State the experiment used to measure acceleration down a slope, and what it measures at each gate.
- 5 Explain why horizontal and vertical motion are independent for a projectile.

- 1 A ball is thrown upward and caught at the same height it was released. Explain why its time up equals its time down.
- 2 Explain why a satellite's orbital speed does not appear in the equations of motion for a projectile on Earth's surface.

Extension

EXTRA PRACTICE

- 1 A projectile is launched at 45° . Explain, without calculating, why this angle gives the maximum range on level ground.
- 2 Sketch the d-t, v-t and a-t graphs for a bouncing ball on one time axis, and check they are consistent.

Extension — EXTRA

EXTRA PRACTICE

- 1 Research the range formula $R = u^2 \sin(2\theta)/g$ and check it against your angled-launch worked example.